

SAAKSHYA **flow**: what runs, why, and what it causes

live:
saakshya.divyanshshukla.com
one call: POST /api/verify

IN · PNG / JPG / PDF

SEVEN STAGES, ONE REQUEST, ONE JSON RESPONSE, FULLY REPRODUCIBLE

OUT · trust level + proof + signed credential

01

INGEST

SHA-256 of bytes.
PDF page rendered
to image at 150 dpi.

file_hash ·
preview.png

02

EXTRACT

RapidOCR (DBNet +
CRNN, ONNX). Regex
typing into fields.

fields · boxes ·
doc_type

03

AUDIT

ELA, noise residual,
copy-move,
metadata, template.
Weighted fusion.

tamper score ·
heatmap

04

CROSS-CHECK

Lookup in issuer
register by roll no /
PAN. Field-by-field
compare.

match · mismatch ·
none

05

DECIDE

Rule table over
tamper score and
cross-check. Human
flag.

L0-L3 ·
needs_human

06

ANCHOR

Salted leaf hash,
Merkle tree, hash-
chained block.

block · root ·
proof

07

ISSUE

Ed25519-signed token,
QR. Verifies offline in
browser.

credential.jwt ·
qr.png

STAGE 05 DECISION RULES

DigiLocker signed, tamper < 0.50 **L3**

Register match, tamper < 0.45 **L2**

Tamper ≥ 0.60, or register mismatch **L0**

Otherwise (no register, clean) **L1**

Tamper 0.40–0.60, or one minor field **human**
off

PROVEN

L3. Issuer's own signature
verifies. Cryptographic
proof, nothing more to
check.

CONFIRMED

L2. Every claim matches the
issuer's register. Forensics
clean.

PLAUSIBLE

L1. Looks clean but no
register exists. We never say
verified here.

REJECTED

L0. Tamper evidence or
register contradiction. Goes
to an officer.

Reading the next pages. For every stage: the technology or algorithm, why we chose it, and the cause-and-effect chain, meaning what input condition triggers what result downstream. Blockchain does one job only (stage 06): prove a verdict existed at a time and was not changed. It does not detect forgery.

01 Ingest

Turn any upload into a fingerprint and a uniform image.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
File fingerprint	SHA-256 over the raw bytes (<code>hashlib</code>)	Collision-resistant identity of the exact file submitted. Standard, fast, court-friendly.	CAUSE same bytes uploaded twice EFFECT identical <code>file_hash</code> , so a resubmission is provable
Type sniff	Magic bytes (<code>%PDF</code>) not the file extension	Extensions lie. A renamed file must not bypass PDF structure checks.	CAUSE PDF detected EFFECT PDF-only forensics and DigiLocker check are enabled in stages 3 and 4
Rasterise	PyMuPDF renders page 1 at 150 dpi to PNG	OCR and pixel forensics need an image. 150 dpi keeps text legible and stays under 2 s on CPU.	CAUSE every input becomes one image EFFECT one code path for scans, photos and PDFs

02 Extract

Read the document into typed claims, then fingerprint the claim itself.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
Text detection	RapidOCR: DBNet finds text regions (differentiable binarisation), returns a box per line	PaddleOCR-grade accuracy through ONNX Runtime. No GPU, no PyTorch, installs in seconds.	CAUSE a box per text line EFFECT boxes drawn on the preview, and reused by template forensics
Text recognition	CRNN with CTC decoding on each box, with a confidence score	Sequence model handles variable-length lines without character segmentation.	CAUSE low confidence on a field EFFECT fuzzy match in stage 4 tolerates it instead of failing
Field typing	Regex + keyword rules: PAN <code>[A-Z]{5}[0-9]{4}[A-Z]</code> , roll no, year, CGPA, DOB; issuer keywords (RGPV, MPBSE, Income Tax)	Deterministic and explainable. A judge or auditor can read the rule. No training data needed.	CAUSE issuer keyword found EFFECT selects which register stage 4 queries
Content hash	Canonical JSON (RFC 8785 style: sorted keys, no spaces, upper-cased values) then SHA-256	We hash the claim, not the file. A rescan changes every byte but not the claim.	CAUSE one digit changed EFFECT different <code>content_hash</code> ; a clean rescan keeps the same one

03 Audit: five forensic signal families

Real tampering touches 0.27–4.17% of pixels (DocForge-Bench 2026). One signal with one threshold misses it, so five independent signals are fused.

SIGNAL (WEIGHT)	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
ELA (0.30)	Error Level Analysis. Re-encode as JPEG q=90, take absolute pixel difference. Compare 32 px blocks against the document's own median and MAD.	A pasted region was compressed a different number of times than its surroundings, so it re-compresses differently. Median/MAD adapts per document instead of a fixed threshold.	CAUSE patch pasted over CGPA EFFECT blocks there deviate, score rises, contour box drawn, heatmap glows on that spot
Noise residual (0.20)	Subtract a median-filtered copy to isolate noise. Standard deviation per 16 px block on flat areas.	Every camera or scanner leaves a uniform noise floor. A foreign patch carries a different one.	CAUSE patch from another source or re-save EFFECT flat blocks with outlier noise are counted into the score
Copy-move (0.20)	ORB keypoints and descriptors, brute-force Hamming self-match, cluster matches sharing one offset > 20 px.	Forgers clone seals, photos and signatures inside the same page. Identical descriptors at a constant offset reveal it. ORB is free and fast.	CAUSE seal duplicated EFFECT dense cluster at one offset, score rises. Repeated glyphs on one text line are ignored on purpose
Metadata / PDF structure (0.15)	pypdf: count of xref sections (incremental updates), Producer string, ModDate vs CreationDate. Pillow: EXIF Software tag.	Editing tools leave file-level traces that pixel edits cannot hide. Immune to AI-generated pixels.	CAUSE Producer = Photoshop / Canva / ilovepdf, or multiple xrefs EFFECT score rises even if the image looks perfect
Template / font (0.15)	Coefficient of variation of OCR box heights per line cluster; issuer keyword expected in header region.	Inserted text rarely matches the form's exact font size and baseline.	CAUSE one value set in a different size EFFECT height variance spikes on that line

FUSION

tamper = 0.30·ELA + 0.20·noise + 0.20·copy
+ 0.15·meta + 0.15·template

(0 ... 1)

Weighted average today. Roadmap: XGBoost on the five scores, calibrated per issuer with conformal prediction for the human-review band.

MEASURED ON OUR SAMPLES			
DOCUMENT	ELA	NOISE	FUSED
rgpv_genuine	0	0	3%
rgpv_tampered	80	75	42% + register says 7.2, doc says 9.1

04 Cross-check

Ask the authority that issued the document whether the claim is true.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
Register lookup	SQL lookup by primary key (roll number or PAN) in the issuer's register. Demo: seeded SQLite tables for RGPV, MPBSE, PAN. Production: APISetu / DigiLocker / NAD.	The only real source of truth is the issuer. Forensics says "edited or not"; the register says "true or not".	CAUSE issuer has no register EFFECT status <code>no_register</code> , level capped at L1 CAUSE key not found EFFECT <code>not_found</code>
Field compare	Normalise (upper-case, strip spaces and punctuation), numeric tolerance on CGPA / percentage, space-insensitive name match.	OCR drops spaces and misreads punctuation. Strict equality would reject genuine documents.	CAUSE doc CGPA 9.1, register 7.2 EFFECT <code>mismatch</code> with both values shown side by side, forces L0
Canonical name	When the name matches, the register's spelling replaces the OCR spelling.	Register is authoritative; fixes "RAHULSHARMA".	CAUSE match EFFECT credential carries the official spelling
Issuer signature	PDF checked for a digital-signature marker (DigiLocker). Production: PKCS#7 verification against issuer certificate.	A valid issuer signature is stronger than any forensic or register check.	CAUSE signed and tamper < 0.50 EFFECT L3 PROVEN

05 Decide

One honest level from two independent kinds of evidence.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
Trust level	Explicit rule table (page 1) over tamper score and cross-check status.	Rules are auditable. An officer can be told exactly why. A black-box classifier cannot be challenged by a citizen.	CAUSE clean forensics but no register EFFECT L1, never a false "verified"
Human flag	<code>needs_human</code> when tamper is 0.40–0.60 or exactly one minor field mismatches.	False positives hurt real people. The AI flags; a person decides.	CAUSE uncertain band EFFECT "Escalated to an officer" with heatmap and register diff in front of them
Confidence	Combination of distance from thresholds and number of matched fields.	Shown on the stamp so officers can prioritise.	CAUSE score near a threshold EFFECT lower confidence printed under the verdict

06 Anchor (the blockchain part)

Prove the verdict existed at this time and has not been changed. Nothing personal goes on chain.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
Salted leaf	<code>leaf = SHA-256(salt content_hash verdict)</code> , salt = 16 random bytes kept off chain.	Without a salt, anyone who knows a student's details could hash them and confirm from the ledger that the student was checked.	CAUSE salt deleted (erasure request) EFFECT leaf still valid on chain but refers to nobody. DPDP-compliant
Merkle tree	Pairwise SHA-256 up to one root (RFC 9162 style, last leaf duplicated when odd). Inclusion proof = sibling hashes with left/right position.	One root commits to any number of verdicts. A verifier checks one verdict with $\log_2(n)$ hashes and sees no other record.	CAUSE 10,000 verdicts in a batch EFFECT 1 ledger write, about 14 hashes per proof
Hash chain	<code>block_hash = SHA-256(index time prev_hash merkle_root)</code> . Demo: SQLite, single node. Target: Hyperledger Fabric on MeitY Vishvasya.	Each block commits to the one before. History cannot be rewritten without recomputing everything after it.	CAUSE one old verdict edited EFFECT its root changes, that block hash changes, every later block breaks. "Verify chain integrity" reports the first broken index
Rejections too	L0 verdicts are anchored exactly like approvals.	A rejection is the record most likely to be disputed or quietly deleted.	CAUSE forger retries with tweaks EFFECT trail of rejections exists and cannot be erased
Revocation	Append <code>SHA-256("REVOKE:" + id)</code> as a new leaf.	Append-only. Nothing is deleted, it is contradicted later.	CAUSE officer revokes EFFECT verifier shows "Revoked"

07 Issue

Give the citizen something reusable that anyone can check without calling us.

STEP	TECH / ALGORITHM	WHY THIS	CAUSE → EFFECT
Sign	Ed25519 (EdDSA) over <code>header.payload</code> , compact JWT form. Payload is a minimal W3C Verifiable Credential with level, issuer, claims, anchor.	64-byte signatures, fast, no parameter pitfalls, supported natively by browsers.	CAUSE any claim altered after issue EFFECT signature fails, verifier shows "NOT valid"
QR	QR code of the verify URL containing the token.	Works on paper printouts and phones.	CAUSE scan EFFECT verifier page opens with the token
Offline verify	Browser WebCrypto imports the public key (JWK) and verifies Ed25519 locally. PWA service worker caches the app and the key.	No dependency on our server being up. No tracking of who verifies whom.	CAUSE no network EFFECT signature still verifies; only revocation status is skipped and says so

END TO END: ONE TAMPERED MARKSHEET, CAUSE TO EFFECT

STAGE	WHAT HAPPENED	EVIDENCE PRODUCED	EFFECT ON THE NEXT STAGE
Forger	Pastes "9.1" over CGPA "7.2", re-saves the image	Patch has different compression and noise	Traces exist before we ever see the file
02 Extract	OCR reads CGPA = 9.1, issuer = RGPV, roll no	Typed fields, content hash of the forged claim	Issuer selects the RGPV register
03 Audit	ELA 80, noise 75 on the patch blocks	Fused tamper 42%, heatmap on the digit	Falls in the 0.40–0.60 band: human flag
04 Cross-check	Register row for that roll number says 7.2	Mismatch: document 9.1 vs register 7.2	Mismatch forces L0 regardless of pixels
05 Decide	Rule: mismatch → L0, uncertain band → officer	REJECTED, escalated	Verdict is fixed and explained
06 Anchor	Salted leaf sealed in a block	Merkle proof, block hash	Rejection cannot be deleted or denied later

The point for judges. Two independent kinds of evidence (pixels and the issuer's register) reach the same conclusion, a person makes the final call, and the record of that call is provable forever without exposing the student's data.